

Problems from take-home exams

1. Let X be a Banach space consisting of continuously differentiable functions on $[0, 1]$ with the property that for each $t \in [0, 1]$ the linear functional

$$l_t(f) = f(t), \quad f \in X$$

is continuous on X . Prove that the linear functionals

$$m_t(f) = f'(t), \quad f \in X$$

are bounded as well and use this to show that the inclusion map $J : X \rightarrow C([0, 1])$

$$Jf = f$$

is compact.

2. a) Let $f \in L^p([-1, 1])$, $1 < p < \infty$, and let Y be the closed subspace of $L^p([-1, 1])$ consisting of even functions. Prove that the closest element in Y to f is $g(x) = \frac{f(x) + f(-x)}{2}$.

b) Is this true in $L^\infty([-1, 1])$ as well?

3. a) Given a Banach space Y and a set A consider the vector space $\ell^\infty(A, Y)$ of all bounded functions $f : A \mapsto Y$ with the norm

$$\|f\|_\infty = \sup_{a \in A} \|f(a)\|_Y.$$

Show that $\ell^\infty(A, Y)$ is a Banach space.

b) Use linear maps from a fixed Banach space into spaces as above to prove that the closed graph theorem implies the uniform boundedness principle.

4. Let $F_1 \subset F_2$ be two nonvoid compact sets in $\mathbb{C}$. Construct a Hilbert space H a bounded linear map M from H into itself and a closed subspace K of H invariant for M ($MK \subset K$) such that

$$\sigma(M) = F_2, \quad \sigma(M|_K) = F_1.$$

5. Consider the operator M_x of multiplication by the independent variable x on $C([0, 1])$, that is

$$M_x f(t) = tf(t)$$

a.e. for all $f \in C([0, 1])$. Here p is fixed in $[1, \infty)$. Find all operators $T \in \mathcal{L}(L^p([0, 1]))$ which commute with M_x , i.e. those which satisfy $TM_x = M_xT$. Solve the same problem for the operator M_x^3 .

6. a) Let (λ_n) be a sequence of complex numbers with $\lim_{n \rightarrow \infty} \lambda_n = 0$ and let (μ_n) be a sequence of positive numbers with $\lim_{n \rightarrow \infty} \mu_n = +\infty$. Find a compact operator T on l^2 with $\lambda_n \in \rho(T)$ for all n and

$$\lim_{n \rightarrow \infty} \frac{\|(\lambda_n I - T)^{-1}\|}{\mu_n} = +\infty.$$

b) Let V be the Volterra operator on $L^2([0, 1])$, i.e. $Vf(x) = \int_0^x f(t)dt$ and recall that $\sigma(V) = \{0\}$. Show that for all $\delta \in (0, 1)$ we have

$$\limsup_{\lambda \rightarrow 0} e^{-\frac{\delta}{|\lambda|}} \|(\lambda I - V)^{-1}\| = +\infty.$$

Hint: The resolvent of V can be explicitly computed.